



Variable Selection in Non-Parametric Regression

Methodological Approach using the Local Linear Estimator, with
Application to Breast Cancer Diagnosis Data



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Introduction

Problem: Modern datasets record hundreds of variables, many of which are redundant or spurious. Retaining such variables reduces model interpretability and leads to overfitting.

Solution: Feature Selection

Problem: Traditional feature selection relies on parametric models, which impose rigid structural assumptions, such as linearity, which can hide more complex relationships.

Solution: Nonparametric Regression

Problem: Nonparametric models suffer from the "curse of dimensionality." High dimensions lead to extreme data sparsity, resulting in high variance, bias, and predictive failure.



How can we robustly identify and select the most important predictive features in high-dimensional data without relying on strict parametric assumptions and combat the curse of dimensionality?

Dataset: Breast Cancer Wisconsin

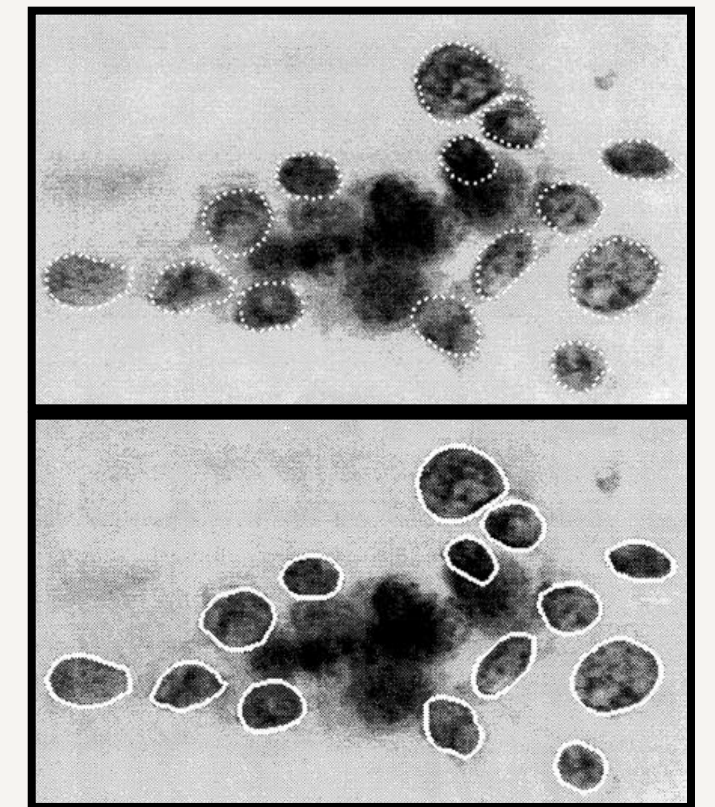
- **Objective:** Diagnose tumors as Malignant ($Y = 1$) or Benign ($Y = 0$) using cell nuclei microscopy.
- **Description:** Features are computed from a digitized image of a Fine Needle Aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image. Obtained from UC Irvine ML Repository.
- **Data:** $n = 569$ patients (357 (62.7%) benign cases and 212 (37.3%) malignant cases).
- **High-Dimensional Feature Space:**
 - $p = 30$ continuous variables.
 - Derived from 10 morphological characteristics (e.g., Radius, Perimeter, etc.).
 - Recorded as Mean, Standard Error, and Worst (maximum) values.

Why not use all 30 features?

- **Redundancy:** Many features are highly correlated (e.g., Radius Mean and Perimeter Mean).
- **Noise:** Capturing non-predictive variance leads to severe overfitting.
- **Interpretability:** Medical diagnostics require simple and explainable models to make better diagnosis.



Solution: Non-parametric Regression



W. Nick Street, W. H. Wolberg, and O. L. Mangasarian "Nuclear feature extraction for breast tumor diagnosis", Proc. SPIE 1905, Biomedical Image Processing and Biomedical Visualization, (29 July 1993);

Non-Parametric Regression

- Nonparametric regression relaxes the structural assumptions of parametric regression.
- Given a sample of independent observations: $(X_1, Y_1), \dots, (X_n, Y_n)$; $X_i \in \mathbb{R}^p, Y_i \in \mathbb{R}$
- $m(x) = \mathbb{E}[Y|X = x]$ represents an unknown, arbitrary regression function.
- Error terms ε_i are independent and centered at zero $\mathbb{E}[\varepsilon_i | X_i] = 0$
- Conditional variance $V(\varepsilon_i | X_i) = \sigma^2(X_i)$

$$Y_i = m(X_i) + \varepsilon_i, \\ i = 1, 2, \dots, n$$

Kernel Density Estimation (KDE):

Estimate the regression function by first estimating the underlying probability density $f(x)$ of our data points using a Kernel function $K(\cdot)$ and bandwidth h .

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

Bandwidth Selection and the Bias-Variance Tradeoff:

- The performance of any kernel estimator is sensitive to the choice of the bandwidth parameter h .
- **Small h :** overly restricted neighborhood; fits to local data points; more statistical noise; low bias-high variance
- **Large h :** expansive neighbourhood; averages over data; washes away structure; low variance- high bias

The Curse of Dimensionality:

- As the number of dimensions increase, the volume of the covariate space expands, leading to extremely sparse local neighbourhoods.
- This sparsity reduces the estimator's convergence rate drastically.

Non-Parametric Regression: Kernel Smoothers

1. Nadaraya-Watson (Local Constant) Estimator:

- Fits a flat constant line to the local data
- $Bias(\hat{m}(x)) \approx h^2 \left(\frac{1}{2} m''(x) + \frac{m'(x)f'(x)}{f(x)} \right) \times \int u^2 K(u) du$
- $Var(\hat{m}(x)) \approx \frac{\sigma^2(x)}{nhf(x)} \int K^2(u) du$

$$\hat{m}_{NW}(x) = \frac{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)}$$

Problem: Suffers from Design Bias (pulled towards regions where density of covariates is high) and severe Boundary Bias (less accuracy towards boundary as data is not present on the other side) $O(h)$.

2. Local Linear Estimator (The Proposed Method):

- Fits a straight, angled line to the local data. (for $q = 1$)
- Find the vector $\beta = (\beta_0, \beta_1, \dots, \beta_q)^T$ that minimizes this following equation and $\hat{m}(x) = \hat{\beta}_0$
- $Bias(\hat{m}(x)) \approx \frac{1}{2} m''(x) \times h^2 \times \int u^2 K(u) du$
- $Var(\hat{m}(x)) \approx \frac{\sigma^2(x)}{nhf(x)} \int K^2(u) du$

$$\min_{\beta} \sum_{i=1}^n \left(Y_i - \sum_{j=0}^q \beta_j (X_i - x)^j \right)^2 K\left(\frac{X_i - x}{h}\right)$$

Advantage: Design-Adaptive (as the design density terms $f(x)$ and $f'(x)$ are not present) and provides Automatic Boundary Correction by maintaining a bias of $O(h^2)$.

Proposed Methodology

1. Initialize

Begin with the full covariate set $\mathbf{A} = \{\mathbf{1}, \dots, \mathbf{p}\}$

2. Parametric Screening

Apply forward-stepwise logistic regression using AIC Penalty ($2k$) to filter redundant variables

3. Restrict Space

Truncate surviving variables into a highly dense subset $\mathbf{S}_0$ to avoid curse of dimensionality

5. Marginal Evaluation

Fit a univariate Local Linear Model $\hat{m}_j(\mathbf{X}_j)$ and compute its **LOOCV MSE**.

4. Nonparametric Optimization

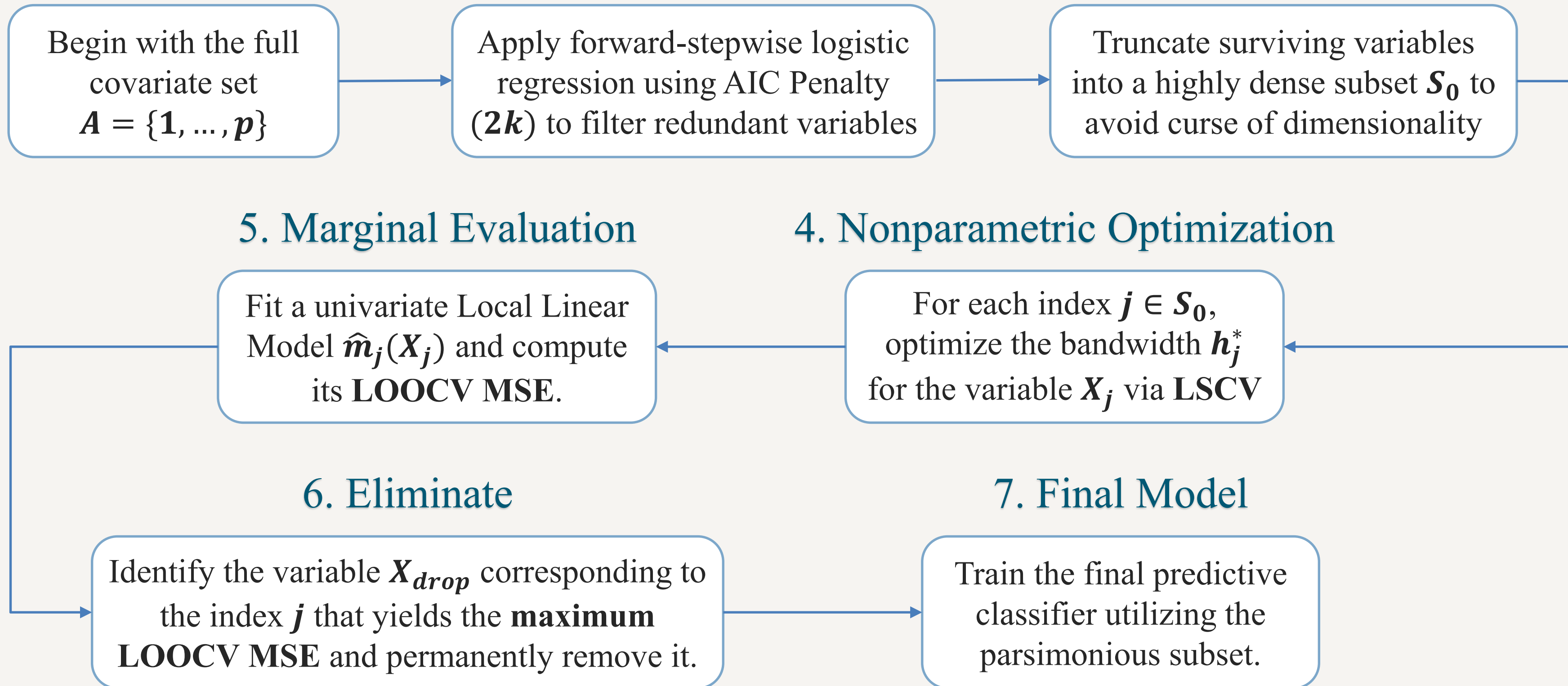
For each index $j \in \mathbf{S}_0$, optimize the bandwidth h_j^* for the variable $\mathbf{X}_j$ via **LSCV**

6. Eliminate

Identify the variable $\mathbf{X}_{drop}$ corresponding to the index j that yields the **maximum LOOCV MSE** and permanently remove it.

7. Final Model

Train the final predictive classifier utilizing the parsimonious subset.



Experimental Verification

Stage 1: Preliminary Parametric Screening; Forward-Stepwise Logistic Regression

AIC over BIC:

- For $n = 455$, BIC penalty is $\ln(455) \approx 6.12$
- Forward-stepwise with BIC truncated the search earlier to only 4 variables.
- On subsequent testing, it leads to an accuracy of 93.68% only.
- AIC has a lighter penalty of only $2k$ which retained 9 variables
- It preserved the essential flexibility needed for the non-parametric stage to locate the true predictive peak.

$$AIC = -2 \ln(\hat{L}) + 2k$$

$$BIC = -2 \ln(\hat{L}) + k \ln(n)$$

$\hat{L}$: maximized likelihood
 k : number of estimated parameters.

Determination of optimal subset size: $k = 6$

- To bypass the *Curse of Dimensionality*, we try to reduce the final subset size to achieve the highest accuracy with the least number of variables.
- The Local Linear estimation and LOOCV feature elimination was run for every $k \in \{3, 4, \dots, 9\}$.
- For $k < 6$: Lack of sufficient information; underfitting
- For $k > 6$: Increasing volume of covariate space; sparse neighbourhoods; more statistical noise

Initial k	Dropped Variable	Final Size	Accuracy
3	smoothness_worst	2	92.11%
4	smoothness_worst	3	93.86%
5	fractal_dimension_se	4	95.61%
6	fractal_dimension_se	5	96.49%
7	fractal_dimension_se	6	93.86%
8	fractal_dimension_se	7	94.74%
9	fractal_dimension_se	8	93.86%

Experimental Verification

Stage 2: Non-parametric Variable Importance; Ranking optimal subset by LOOCV MSE

Focus Covariate Evaluation:

- For the $k = 6$ optimal subset of covariates, declare X_j as the focus.
- Estimate its marginal relationship with the response variable Y using Local Linear Estimation
- For each X_j , the bandwidth parameter h_j is optimally selected via Data-Driven Least Squares Cross Validation:

$$\hat{Y}_j = \hat{\mathbb{E}}[Y|X_j] = \hat{m}_j(X_j; h_j)$$

$$h_j^* = \arg \min_h \text{LSCV}(h)$$

Error Evaluation:

- After obtaining h_j^* , we calculate the predictive Mean Squared Error of the covariate using Leave-One-Out Cross Validation (LOOCV) MSE.
- Here, $\hat{m}_{-i}(X_{ij}; h_j^*)$ denotes the Local Linear estimate at X_{ij} computed using the data excluding the i -th observation. The MSE is the average squared prediction error across all left-out folds.
- Lower MSE indicates higher marginal predictive utility i.e. the variable predicts Y well in isolation

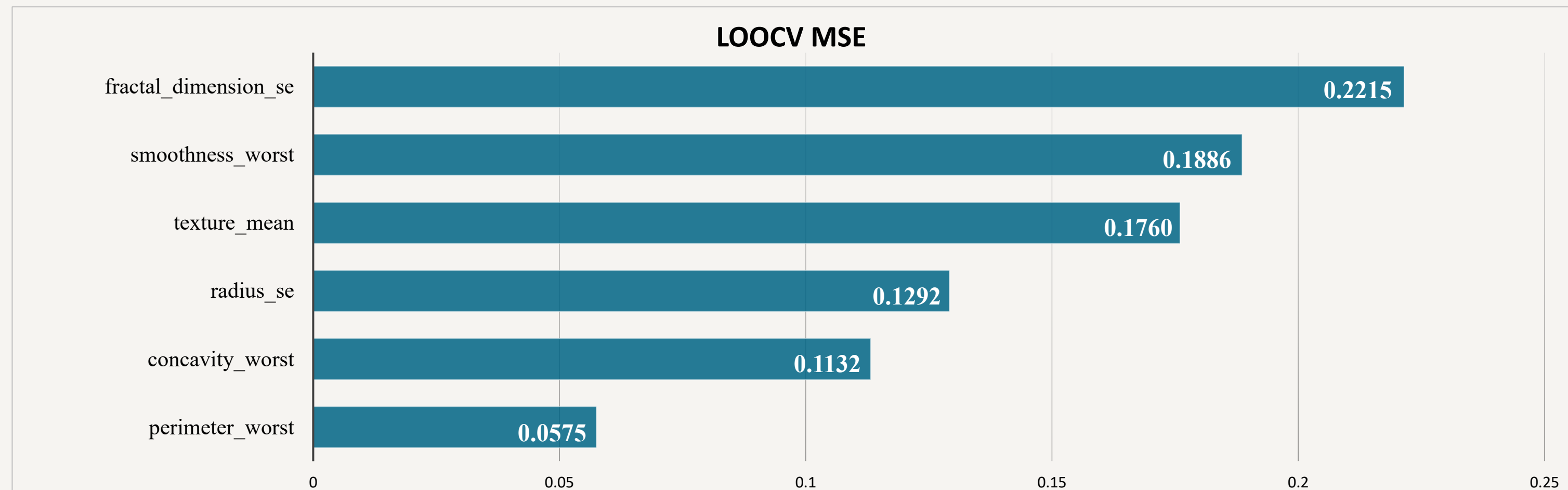
$$\text{MSE}_{CV}(j) = \frac{1}{n} \sum_{i=1}^n \left(Y_i - \hat{m}_{-i}(X_{ij}; h_j^*) \right)^2$$

Experimental Verification

Stage 2: Non-parametric Variable Importance; Ranking optimal subset by LOOCV MSE

Backward Elimination:

- Rank the variables according to their LOOCV MSE : $MSE(j_1) < MSE(j_2) < \dots < MSE(j_{k_0})$
- *fractal_dimension_se* has maximum error. It is dropped as it contains the least unique predictive information.
- *perimeter_worst* has the lowest error, making it the most structurally dominant variable



Final Results

Parsimonious Final Model

5 optimally selected variables after dropping *fractal_dimension_se*

Final logistic regression classifier was trained exclusively on the 5-variable subset and evaluated on the unseen testing data

Test Accuracy:
96.49%

Negative Control Model

A Negative Control model was trained using the 25 discarded variables

The control model triggered mathematical convergence warnings due to extreme multicollinearity.

Test Accuracy:
94.74%

Conclusion

- **Two-Stage Screening Framework:** Successfully combined the computational efficiency of parametric screening with the flexibility of nonparametric smoothing.
- **Parsimony Outperforms Complexity:** A strictly reduced subset of just 5 features retained the vast majority of structural information, yielding **96.49% accuracy**.
- **Mathematical Reliability:** Exclusive use of the Local Linear estimator guaranteed feature selection remained strictly robust against boundary bias and dimensional sparsity.

Limitations and Future Work

- **Marginal vs. Joint Evaluation:** Extending to evaluate additive models (e.g., Generalized Additive Models) which could capture interaction effects between covariates.
- **Nested K-fold Cross-Validation for Hyperparameters:** Instead of a strict 80/20 train-test partition; to provide a more robust, unbiased estimate of generalization error and to rigorously tune the initial subset size (k) without data leakage.
- **Statistical Significance of MSE Rankings:** Point estimates of MSE contain sample variance. Implementing bootstrap resampling to generate 95% confidence intervals around the LOOCV errors would allow formal hypothesis testing.

Appendix: Local Linear Estimator

Local polynomial estimator: Find vector $\beta = (\beta_0, \beta_1, \dots, \beta_q)^T$ that minimizes

$$\min_{\beta} \sum_{i=1}^n \left(Y_i - \sum_{j=0}^q \beta_j (X_i - x)^j \right)^2 K \left(\frac{X_i - x}{h} \right)$$

Then $\hat{m} = \widehat{\beta}_0(x)$.

For **local linear estimator**, $q = 1$.

Then, the minimizer is:

$$\hat{\beta} = \left(\widehat{\beta}_0(x), \widehat{\beta}_1(x) \right)^T = (\mathbb{X}^T W \mathbb{X})^{-1} \mathbb{X}^T W \mathbb{Y}$$

$$K_h(u) = \frac{1}{h} K(u/h)$$

$$\mathbb{Y} = (Y_1, \dots, Y_n)$$

$$W = \begin{pmatrix} K_h(x - X_1) & 0 & \dots & 0 \\ 0 & K_h(x - X_2) & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & K_h(x - X_n) \end{pmatrix}$$

$$\mathbb{X} = \begin{pmatrix} 1 & X_1 - x \\ 1 & X_2 - x \\ \vdots & \vdots \\ 1 & X_n - x \end{pmatrix}$$

Appendix: Nadaraya-Watson Estimator

The estimator is defined as:

$$\hat{m}_{NW}(x) = \frac{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) y_i}{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)}$$

Derivation:

$$E(Y | X = x) = \int y f(y|x) dy = \int y \frac{f(x, y)}{f(x)}$$

The joint distributions $f(x, y)$ and $f(x)$ are estimated using kernel density estimation with kernel K .

$$\hat{f}(x, y) = \frac{1}{n} \sum_{i=1}^n K_h(x - x_i) K_h(y - y_i)$$

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^n K_h(x - x_i)$$

where $K_h(u) = \frac{1}{h} K(u/h)$

$$\hat{E}(Y|X = x) = \int y \frac{\hat{f}(x, y)}{\hat{f}(x)} dy$$

$$= \int y \frac{\sum_{i=1}^n K_h(x - x_i) K_h(y - y_i)}{\sum_{i=1}^n K_h(x - x_i)} dy$$

$$= \frac{\sum_{i=1}^n K_h(x - x_i) \int y K_h(y - y_i) dy}{\sum_{i=1}^n K_h(x - x_i)}$$

$$= \frac{\sum_{i=1}^n K_h(x - x_i) y_i}{\sum_{i=1}^n K_h(x - x_i)}$$

Appendix

Least Squares Cross Validation:

To prevent $h \rightarrow 0$, LSCV uses Leave-One-Out estimator. For every point i , we hide (X_i, Y_i) and fit the curve using the remaining $n - 1$ points, denoted as $\hat{m}_{-i}(X_i)$.

The LSCV objective function is:

$$LSCV(h) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{m}_{-i}(X_i; h))^2$$

The optimal bandwidth is $h_j^* = \arg \min_h LSCV(h)$

Leave One Out Cross Validation Error:

It is the unbiased generalization error used to rank the variables.

It is calculated using the optimal bandwidth h_j^*

$$MSE_{CV}(j) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{m}_{-i}(X_{ij}; h_j^*))^2$$

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Thank You

